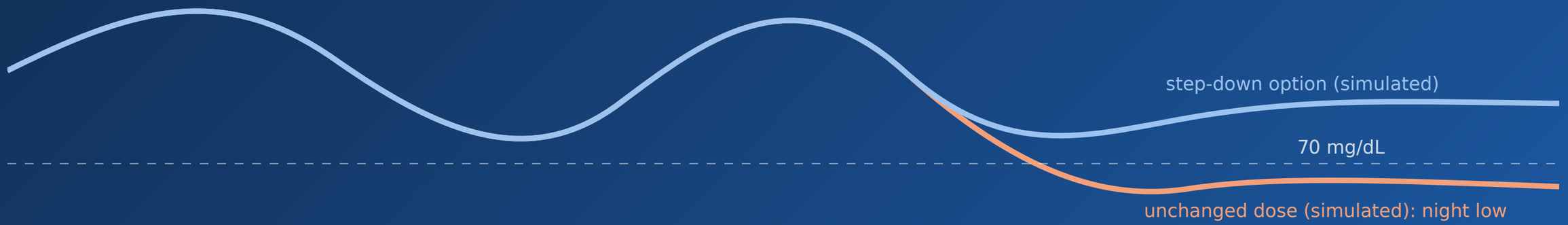
HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026 · PHASE 1

Santulan

A research prototype for clinician-led diabetes dose review: a digital twin that brings a patient's body changes and CGM history together, with forecasts evaluated on real held-out patients and dose scenarios whose assumptions are stated



Team Heartificial Intelligence · Shubham Sunil Patil, Adarsh Nayan · IIM Mumbai

Code, data and results on GitHub: Heartificial-Intelligence_IIM-Mumbai

THE PROBLEM

Millions of Indians take the two diabetes drug classes that can push sugar dangerously low

Sulfonylureas and insulin are India's workhorse drugs. When the body changes, the same dose can become too strong

10.1 cr

Indian adults with diabetes

ICMR-INDIAB 2023

2.5 cr

take a sulfonylurea or insulin: about half are treated, half of those on these drugs

LANDMARC, 6,234 patients

25 lakh

severe lows a year: our estimate of the problem's size (range 13 to 48 lakh), not a prototype result

Meta-analysis of 5.3 lakh people: 5% a year on SU, 21% on insulin

48 of 164

drug-reaction admissions at KEM Hospital, Mumbai, were sulfonylurea reactions

JCDR 2022

Lows are often silent

In older adults wearing a sensor, 80 to 100% of lows went unnoticed. HbA1c can read "well controlled" while nights go low.

Systematic review, UEA

Lows harm

Severe lows are linked to dementia, and in the LEADER trial to a higher risk of death.

JAMA 2009; LEADER

Lows are common

8.9% of insulin-treated Type 2 patients had a severe low in just four weeks; South-East Asia, including India, was among the highest.

Global HAT study, 2016

Sources in the README

WHAT SANTULAN IS

A working twin for the dose review, with every part labelled by how far it is proven

A physiology core fitted to each person, population models evaluated on held-out patients, stated review rules and illustrative scenarios

Calibrate IMPLEMENTED

A fixed 5-day CGM window fits the person's physiology core and freezes their sensor offset from calibration fingersticks.

Forecast EVALUATED

30, 60 and 120-minute glucose forecasts on real held-out patients, against honest baselines.

Low alert EVALUATED

A simple rule on the corrected reading, because no model beat it at the same alert budget.

Review rules RULES

GLP-1 start, weight loss, planned fast, eGFR below 60; G6PD, CYP2C9 and age notes, each with its label note.

Dose scenarios ILLUSTRATIVE

Step-down options compared on the person's core, with assumptions on sliders and their sensitivity shown.

Sensor-off EXPERIMENTAL

Glucose estimation from doses, meals and fingersticks; tracks the level, cannot detect lows.

Live replay steps a real held-out patient through time with the same code as the evaluation. Smartwatch data are display only; FHIR records are example shapes; quarterly sensing is proposed. **The doctor decides; the software never changes a dose.**

THE TWIN IN ACTION

Live replay: a real held-out patient, stepped through time with the evaluation's own code

Only records up to the replay clock are used; the forecast model was trained without this patient

Replay clock 21 Nov 2020 02:52 · **Calibration** 10 Nov 11:22 to 15 Nov 11:22 · **Model** fold 4 (this patient excluded from training) · **Alert rule** corrected reading at or below 79.0 mg/dL

Forecast from the last reading, using records up to the replay clock.

Last reading (corrected)	Forecast +30 min	Forecast +60 min	Forecast +120 min	Alert now
76 mg/dL	82	94	98	YES

Low alert: the corrected reading 76 mg/dL is at or below 79.0 (cooldown 60 minutes after each emitted alert). In the retrospective evaluation this simple rule caught at least as many lows as the forecast models at the same alert budget, so it is the alert used here; the forecasts are shown for context.

Scoring, revealed: the evaluation's alert log has this period's latest alert at 21 Nov 02:52 too, scored matched: a low began within the next hour (low began 03:52, 60 minutes after the alert).



Dotted orange lines: alerts emitted by the shared alert policy up to the clock.

02:52, an hour before a night low

76 mg/dL is at or below the fold's 79 threshold, so the simple rule alerts; the forecasts expected a rise. The low began at 03:52. An earlier low was missed.

What updates when

- Each step adds new CGM readings, doses and meals
- Offset and core are fitted once on calibration, not refitted online

Checks you can see

- Forecasts equal the evaluation's stored predictions
- Stale or missing CGM shows "unavailable", not a guess
- A synthetic GLP-1 start changes the review rule, not the physiology

twin/replay.py; tests/test_replay.py

WHY NOW

In 2026 the body changes faster than the prescription

Four common changes can make a previously right sulfonylurea or insulin dose too strong, and doses rarely come down

GLP-1 start

17 to 24%

have lows on semaglutide plus a sulfonylurea, against 2 to 4% alone; the label advises considering a lower dose. Generic semaglutide reached India in March 2026.

Drug label rates; South First

Weight loss

5%+

raises insulin sensitivity, so the same tablet can lower sugar further. Wellness success is exactly when doses may need review.

Project Khushi pilot: 59% lost weight

Fasting

Fast days

Meals move or vanish on fasting days; doses often do not.

JSHD 2015; IJEM 2017

Kidney function

eGFR < 60

The glimepiride label advises a 1 mg start and conservative titration; its metabolites accumulate as kidney function falls.

US glimepiride label

9.4 to 16.9%

Among 89,987 US adults aged 65 and over, 48% were already below their sugar goal, yet only 9.4% (2019) to 16.9% (2023) had treatment reduced. Clinicians need structured support to decide when a dose should come down.

JAMA Internal Medicine, via Drug Topics; US glimepiride label

THE GAP

Low prediction already exists; structured support for the dose decision does not

Santulan aims at the medication review: which body changes happened, what the CGM shows, and how step-down options compare

Prediction is already a product

Roche's SmartGuide app predicts lows 30 minutes ahead, glucose 2 hours ahead and overnight risk on a sensor worn continuously. On sale in the Gulf; we found no India launch.

Roche UK; BioSpectrum Asia

Whole-body twin programmes exist

Twin Health runs a paid remission programme first trialled in India. It is not a clinician's dose-review tool for sulfonylurea and insulin users.

ADA 2022 report; NEJM Catalyst via pharmaphorum

Where Santulan fits

Not another low predictor: support for the review a physician does when a body change makes a dose questionable, with the evidence behind each part stated.

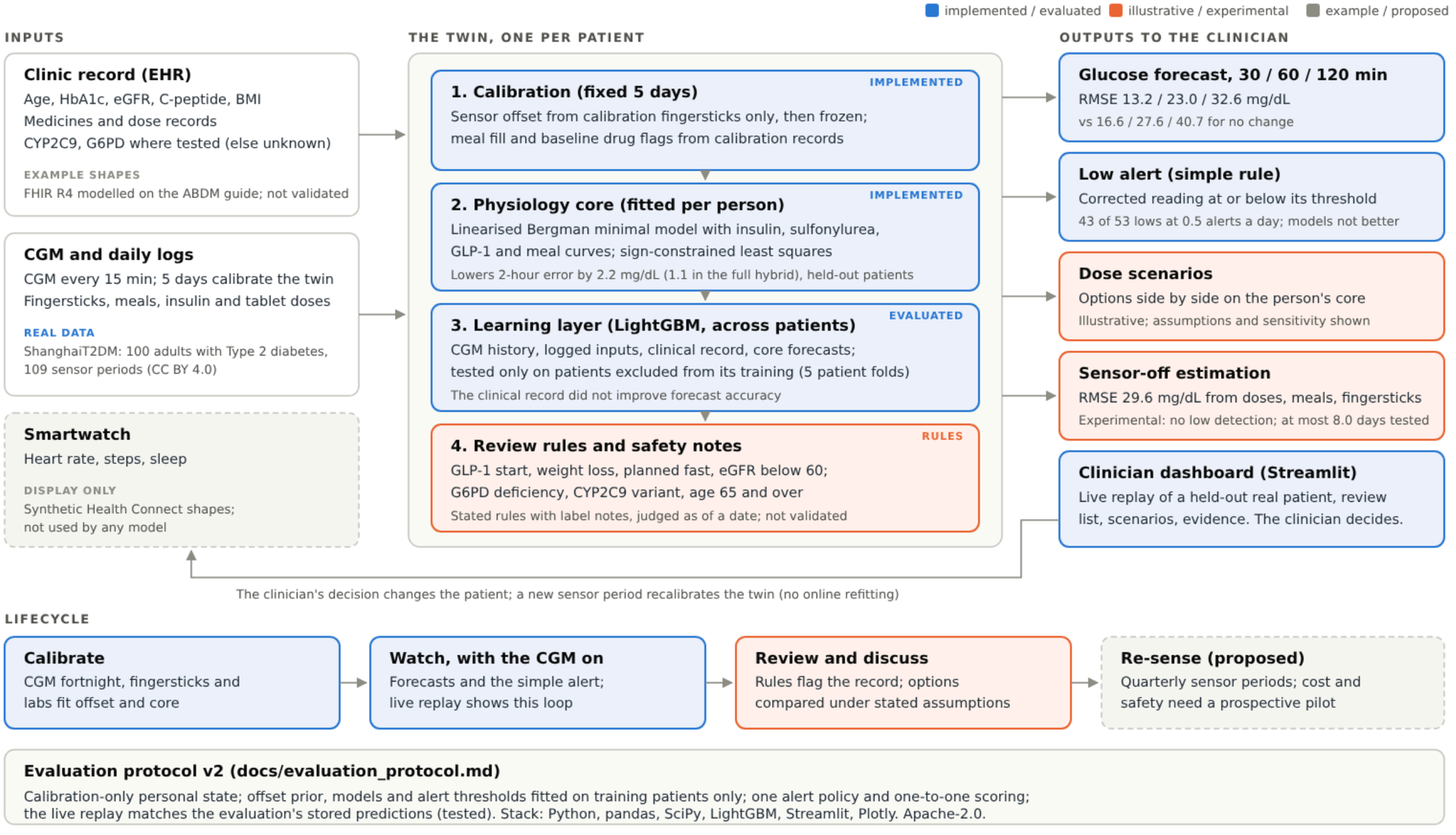
What 28 public challenge entries do (keyword scan, 14 Oct)

Type 2 glucose twins	**12**
What-if simulators	**18**
Watch for GLP-1, weight-loss or fasting changes	**0**
Test the twin with the sensor off	**0**
Use CYP2C9 or G6PD for sulfonylureas	**0**
Mention sensor artefacts at all	**1**

Indicative counts from a keyword scan of entries public on one day. Santulan's own results on these points are shown in this deck, including where they are negative.

Santulan: a research prototype for clinician-led diabetes dose review

Architecture and data flow. Each part is labelled by its status. The clinician decides; the software never changes a dose.



LIFECYCLE

Calibrate
CGM fortnight, fingersticks and labs fit offset and core

Watch, with the CGM on
Forecasts and the simple alert; live replay shows this loop

Review and discuss
Rules flag the record; options compared under stated assumptions

Re-sense (proposed)
Quarterly sensor periods; cost and safety need a prospective pilot

Evaluation protocol v2 (docs/evaluation_protocol.md)

Calibration-only personal state; offset prior, models and alert thresholds fitted on training patients only; one alert policy and one-to-one scoring; the live replay matches the evaluation's stored predictions (tested). Stack: Python, pandas, SciPy, LightGBM, Streamlit, Plotly. Apache-2.0.

HOW WE EVALUATE

An audit found our first evaluation let later data in; version 2 is leak-free

76 real patients, 85 sensor periods, 538 monitored days (ShanghaiT2DM, open data). Protocol frozen before running; tests enforce it

What version 1 got wrong

- Sensor offset used fingersticks from later days
- The offset prior included the patients being tested
- Missing meals and the calibration length looked at the whole period
- Sensor-off mode started from a hidden CGM reading
- Alarm hits and false alarms were scored by different rules

Its results are kept, labelled superseded, and not quoted.

What version 2 does

- Fixed 5-day calibration per person; offset, meals and core from those days only
- Population models, offset prior and alert thresholds fitted on training patients only (5 patient folds)
- One alert policy; one-to-one matching; false alerts only with observed follow-up
- Honest baselines: no change, recent trend, physiology only, ML without physiology
- Tests check that later data cannot change an earlier output

5 days
fixed calibration per person, after a 24-hour warm-up

5 folds
of patients; population models never saw the patient, who is personalised only by their own 5 days

53 lows
CGM-defined episodes in 20 held-out patients

Retrospective
same patients as version 1: development evidence, not independent validation

docs/evaluation_protocol.md; results/manifest.json

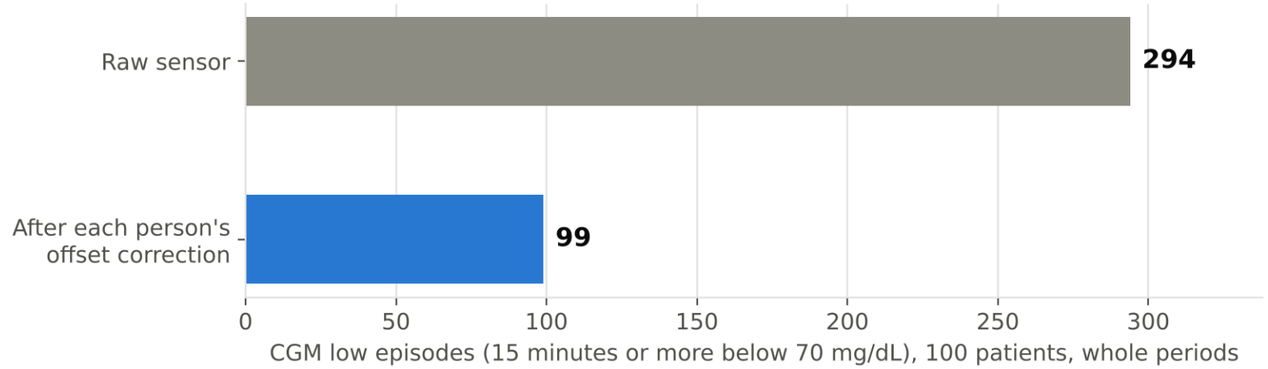
REAL DATA, DESCRIPTIVE CHECK

The sensor read 12 mg/dL low; a whole-period offset correction changes low episodes from 294 to 99

ShanghaiT2DM: 100 adults, 109 sensor periods, 112,287 readings, 3,262 fingersticks

A whole-period offset correction cuts sensor low episodes from 294 to 99

Descriptive check: each person's offset from all of their fingersticks (the twin itself uses calibration fingersticks only). The sensor read 12 mg/dL below fingersticks on average; 7 of 53 paired sensor readings below 70 (fingerstick within 8 minutes) were confirmed below 70. Fingersticks are sparse, so true lows are not established.



12 mg/dL
sensor below fingersticks, across 2,883 pairs

7 of 53
paired sensor readings below 70 (fingerstick within 8 minutes) were confirmed below 70

What the twin does

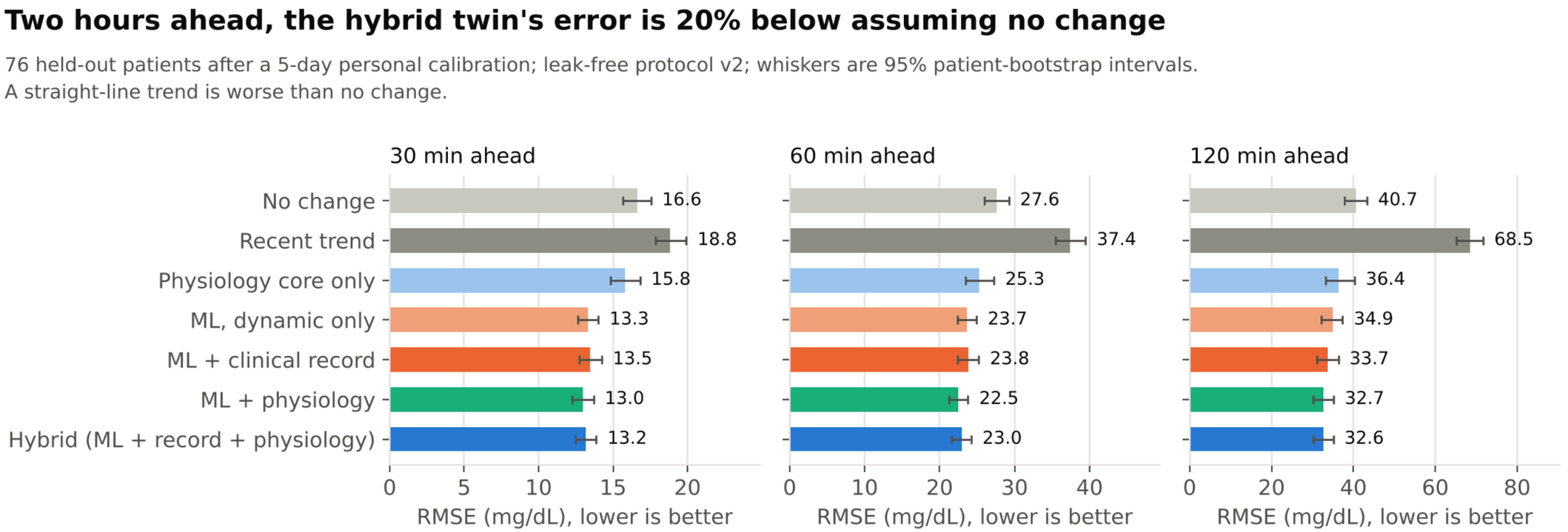
- Drops each sensor's first 24 hours
- Learns the offset from calibration fingersticks only, then freezes it
- Counts a low only after 15 minutes below 70

Descriptive and whole-period. Sparse fingersticks do not establish true lows. A US hospital study found a 12.8 mg/dL bias for this sensor family (Diabetes Care 2020).

REAL DATA, FORECASTS

Two hours ahead, the hybrid twin's error is 20% below assuming no change

Held-out patients after a 5-day personal calibration; every method scored on the same origins



32.6
mg/dL hybrid error at 2 hours (95% CI 30.2 to 35.1)

40.7
mg/dL for "no change"; 36.4 for the physiology core alone

13.2 / 23.0
mg/dL hybrid error at 30 and 60 minutes

68.5
mg/dL for a straight-line trend: worse than no change

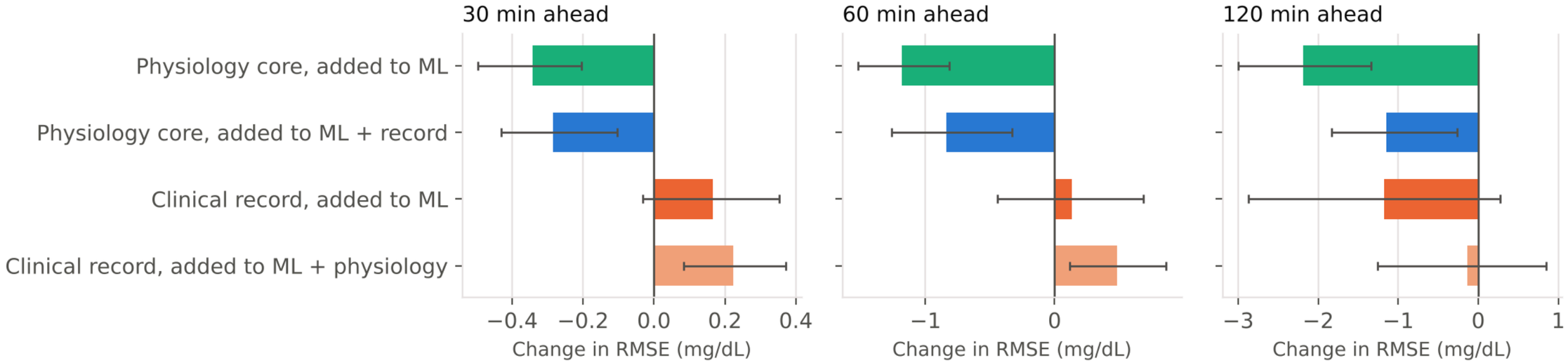
WHAT EACH INGREDIENT ADDS

The physiology core improves forecasts; the clinical record does not

Paired comparisons on identical patients and origins, with patient-level bootstrap intervals

The physiology core improves forecasts; the clinical record does not

Paired comparisons on identical origins and patients, 95% patient-bootstrap intervals; left of zero means the ingredient lowers error.
The clinical record's role is the review triggers, not forecast accuracy.



Physiology core: adding it lowers 2-hour error by 2.2 mg/dL to machine learning without the record, and by 1.1 mg/dL in the full hybrid; both intervals exclude zero. This is the twin's measured contribution.

Clinical record: once the twin is calibrated, it makes forecasts slightly worse at 30 and 60 minutes and no different at 2 hours. Its job is the review rules: medicines, eGFR, weight, genetics and fasting plans decide who is flagged.

results/evaluation.json: paired_differences

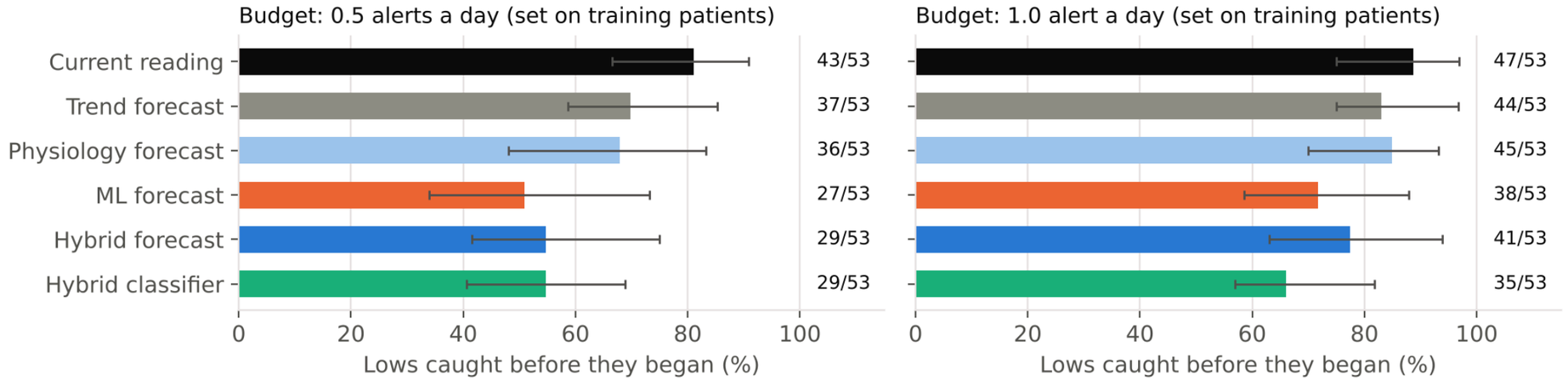
REAL DATA, LOW ALERTS

At the same alert budget, a simple rule caught 43 of 53 lows; the hybrid caught 29 of 53

53 CGM-defined episodes in 20 held-out patients; thresholds set on training patients; one-to-one matching, 60-minute window and cooldown

At the same alert budget, a simple rule on the current reading caught the most lows

53 CGM-defined low episodes in 20 held-out patients; one-to-one matching, warning window 60 minutes, 60-minute cooldown. Intervals overlap widely: no model is shown to improve low alerts.



So the prototype alerts on the simple rule

Corrected reading at or below 77 to 80 mg/dL (set per fold on training patients), 60-minute cooldown; forecasts shown for context.

Mixed and fragile

On raw CGM the hybrid caught 88 of 149 against 73. Small pipeline fixes moved the hybrid from 36 to 29 of 53; the simple rule stayed at 43.

Rare events

Classifier AUROC 0.83 but AUPRC 0.11 at 0.39% prevalence. Episodes are sensor-defined, not severe clinical events.

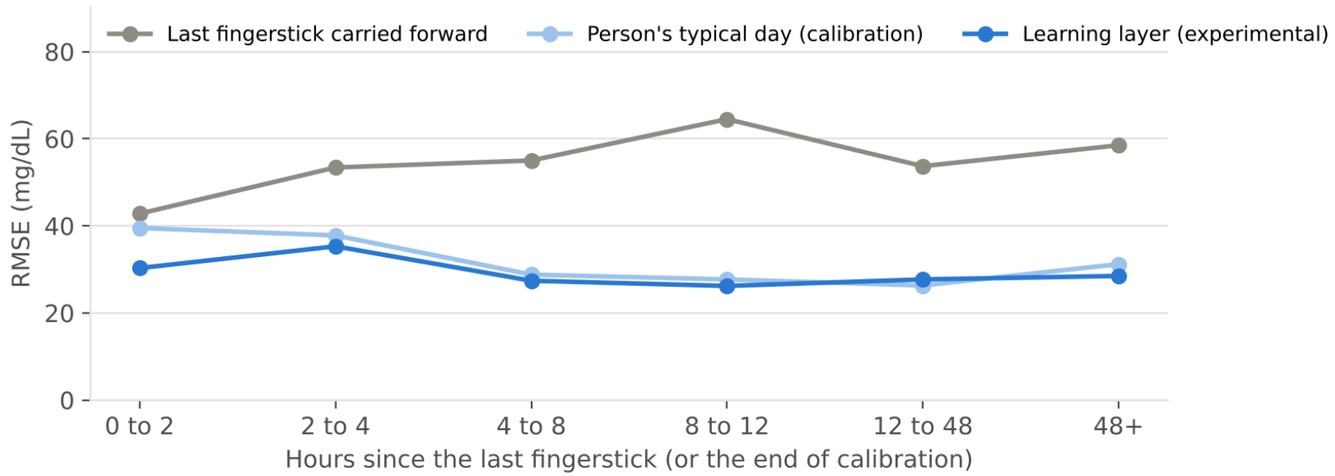
EXPERIMENTAL: SENSOR-OFF ESTIMATION

With the CGM off, estimates track the level (29.6 mg/dL) but not the lows

After calibration the core sees only doses, meals and fingersticks; hidden CGM is used only for scoring

Sensor off: estimates track the level, not the lows

7,600 hours in 46 patients, at most 8.0 days per period. RMSE 29.6 vs 33.6 (typical day) and 52.2 (last fingerstick); only 0.5% of readings below 70 were estimated below 80. Acute low alerts are not available in this mode.



29.6

mg/dL RMSE, against 33.6 for the person's typical day and 52.2 for the last fingerstick

1 of 213

readings below 70 were estimated below 80

Scope

- Acute low alerts are not available without a CGM
- At most 8.0 days per period tested, in 46 patients
- The twin alone drifts after 2 days; the learning layer leans on the person's typical day
- Quarterly sensing between periods is a proposal for a pilot, not a result

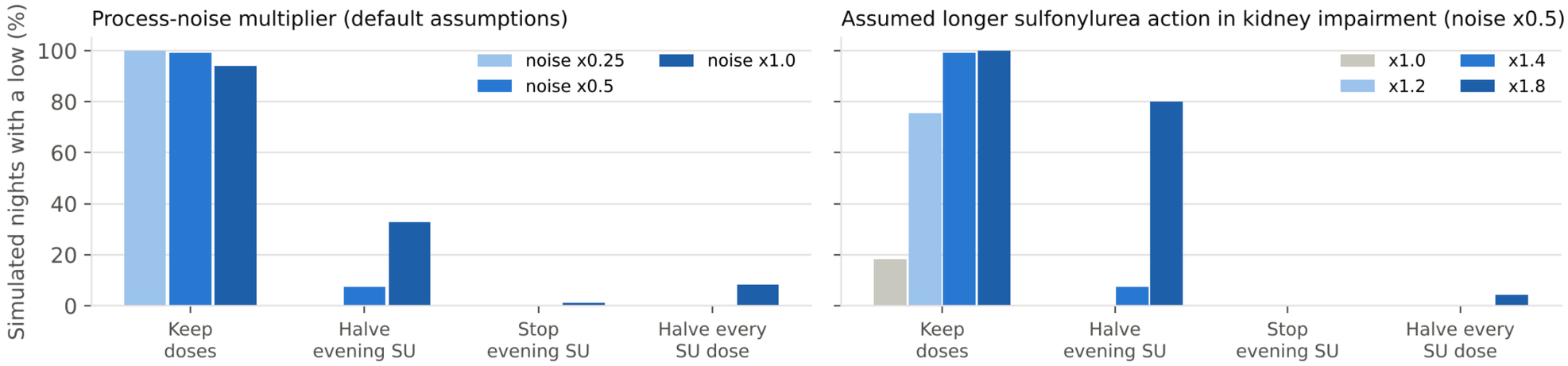
ILLUSTRATIVE DOSE SCENARIOS

Meena's scenario ranks the options, but its percentages swing with the assumptions

Meena, 62 (synthetic): glimepiride-metformin twice daily, semaglutide for 6 weeks, 5% weight lost, eGFR 72 to 58; physiology copied from one real patient

Meena's scenario: the order holds, but the percentages swing with the assumptions

Synthetic persona; physiology copied from one real patient's fitted core. 300 simulations x 14 nights per option. Comparative and uncalibrated: these are not risk estimates or treatment effects.



Default assumptions: keep doses 99% of simulated nights with a low; halve the evening dose 7.5%; stop it 0%.

But: halving the evening dose ranges from 0% to 33% across noise settings and reaches 80% under a stronger kidney assumption; without it, unchanged doses give 18%.

Regimen: twice-daily glimepiride-metformin combinations are common in India (7,058 patients, 347 sites); the US label specifies once daily. The doctor decides.

results/scenario_sensitivity.json

HEALTHCARE USE CASE

A medication-review tool for the clinics that start GLP-1s and see older patients

We claim no prevented lows or savings; a prospective pilot would measure them

The physician's workflow

- A review rule flags the record, with its reason
- CGM history, forecasts and the simple alert
- Step-down options compared under stated assumptions
- The physician decides; a new sensor period recalibrates

Where it could be piloted

- Endocrinology and Your Personal Physician clinic, Koramangala
- The planned obesity clinic, where GLP-1 starts happen
- Healthy Ageing, where older adults on sulfonylureas are
- Project Khushi: wellness success is when doses may need review

What a pilot must show

- Severe lows needing assistance
- CGM time below 70 mg/dL
- Dose reductions made and their outcomes
- False-alert burden and clinician acceptance
- Body-change effects fitted on Indian patients

Fit with Happiest Health: labs from Happiest Diagnostics could seed the calibration; records modelled on ABDM FHIR shapes could carry consented data later; "Probe. Predict. Prevent.": the twin supports the Predict.

Happiest Health clinic launch, July 2026; Project Khushi report

DATA, SAFETY, LIMITS AND REPRODUCIBILITY

Open data, stated limits, and every number regenerated by code

What judges can rerun, and what a pilot must still prove

Data

- ShanghaiT2DM: 100 real adults, CC BY 4.0 (public mirror matching the paper)
- 24 synthetic Indian patients for the demo; never used to grade a model
- Smartwatch data display only; FHIR files are example shapes

Safety

- Decision support only; no dose is changed by software
- Rules cite label notes; scenarios labelled illustrative
- No personal data; consent under the DPDP Rules needed for real use
- Apache-2.0

Limits

- 53 CGM-defined lows in 20 patients: wide intervals
- Chinese adults; Indian calibration needed
- Retrospective development evidence
- Scenario effects are assumptions

Rerun everything

```
pip install -r requirements-lock.txt
python scripts/run_all.py    # every result, summary and figure, about 15 to 20 minutes
python -m pytest -q tests    # causality, alert matching, replay parity, document consistency
streamlit run app/dashboard.py
```

SANTULAN

A working, honestly evaluated twin for the dose review; a clinical pilot is the next test

20%

lower 2-hour forecast error than no change (32.6 vs 40.7 mg/dL), on 76 held-out patients

2.2 mg/dL

the physiology core's 2-hour gain without the clinical record (1.1 in the full hybrid); the record drives the review rules instead

Leak-free

protocol, fair baselines, negative results published, and a live replay that matches the evaluation's stored predictions

Shown on real data

Forecasts and the simple low alert, on 76 held-out patients; the replay matches them

Not yet proven

Dose scenarios and review rules are illustrative; sensor-off cannot detect lows; no outcome is claimed

What a pilot needs

Consented Indian CGM and clinic data, a partner clinic, and endpoints fixed in advance

Team Heartificial Intelligence · Shubham Sunil Patil, Adarsh Nayan · IIM Mumbai

GitHub: Heartificial-Intelligence_IIM-Mumbai